How Far Does a Litter Detector Travel? Cross-Domain Urban Waste Detection Across Handheld, Dashcam and Drone Imagery

Jairzinho Santos
Universidad Nacional de Ingeniería
Lima, Peru
Computer Vision, MSc in Artificial Intelligence
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***Abstract*—Automated litter monitoring is usually evaluated within a single imaging domain, yet real deployments mix handheld reports, vehicle cameras and drones. We study how much urban waste detectors degrade across three public domains TACO (handheld), RoLID-11K (dashcam) and UAVVaste (drone) under a single audited evaluation protocol. Our curation uncovered seven defects in the released datasets, most notably video-level leakage in RoLID-11K's official splits, where 58.2% of test images share a source video with training; a model trained with that overlap scores +21% AP$_{50}$ on the same test set. We compare Faster R-CNN, an FCN segmenter and CAM-based interpretability against a YOLOv11 reference, and measure a cross-domain degradation of 82% mAP$_{50}$, dominated by object scale (median 22 px in dashcam imagery, below the smallest FPN anchor). Raising input resolution to 1024 px recovers up to 8.3 AP$_{50}$ and 5.8 AP$_S$ points. Code, weights and audited splits are released.**

***Index Terms*—litter detection, cross-domain evaluation, dataset auditing, small objects**

I. Introduction

Uncollected urban waste harms public health, tourism and municipal budgets, and its monitoring still relies on manual inspection with limited coverage. Computer vision promises continuous, low-cost monitoring, and recent work has produced public datasets for litter detection in complementary settings: street-level handheld photographs (TACO [1]), forward-facing dashcams (RoLID-11K [2]), drone imagery (UAVVaste [3]) and waste-truck fisheye cameras (StreetView-Waste [4]). Each benchmark, however, is evaluated *in-domain*; whether a detector trained on one viewpoint remains useful on another the situation any city deploying mixed sensors faces has not been measured.

This paper asks: *how much does urban litter detection degrade across imaging domains, and what matters most to close the gap architecture, input resolution, or training data hygiene?* Answering it required more than training detectors. Auditing the three datasets surfaced seven defects (Section III), including video-level leakage in RoLID-11K's official splits that inflates published baselines, basename collisions and inconsistent EXIF annotation frames in TACO and UAVVaste. We therefore built an audited curation pipeline with frozen, group-aware splits, and evaluate every model against the same ground truths.

Our contributions are: (i) a documented audit of three public litter datasets, with corrected and released splits; (ii) the first cross-domain evaluation matrix across handheld, dashcam and drone litter imagery; (iii) a quantification of the leakage inflation in RoLID-11K's official protocol (+21% AP$_{50}$ from scene memorization); (iv) a controlled comparison of classical fine-tuned detectors, segmentation and interpretability techniques against a modern real-time reference, including a small-object resolution ablation and a class-granularity (6 vs. 5 material classes) study.

II. Related Work

Litter datasets. TACO [1] provides 1,500 handheld images with 60 litter categories and segmentation polygons. RoLID-11K [2] contributes 11.5k dashcam frames with a single *litter* class and pronounced small-object statistics. UAVVaste [3] covers low-altitude drone views. StreetView-Waste [4] targets waste-truck fisheye cameras; its download endpoints were unavailable during this work. Detect-waste [5] harmonized several of these sources; ZeroWaste [6] addresses industrial sorting, a domain we deliberately exclude.

Detection under constraints. TrashDet [7] searches TinyML detectors on TACO, reaching 19.5 mAP$_{50}$ on a five-class subset evidence that absolute scores in this problem are structurally low, which motivates our focus on *relative* cross-domain quantities. RoLID's authors benchmark modern detectors, with transformer heads (CO-DETR) leading YOLO variants by over 20 AP$_{50}$ points, consistent with the extreme small-object regime.

Methods. We build on Faster R-CNN [8] and its RPN, fully convolutional segmentation [9], class-activation interpretability [10], [11], [12], YOLOv11 [13] as the real-time reference, TIDE [14] for error decomposition and iterative stratification [15] for group-aware splitting.

III. Datasets, Audit and Curation

We ingest the three in-scope datasets through a medallion pipeline (raw→bronze→silver→gold) with per-layer manifests

TABLE I

THE THREE DOMAINS AFTER CURATION. MEDIAN OBJECT SIDE MEASURED ON THE ORIGINAL IMAGES; AT THE 640-PX TRAINING RESOLUTION THESE BECOME $\sim$32, $\sim$7 AND $\sim$12 PX RESPECTIVELY.

Dataset	Viewpoint	Images	Median obj.	% small
TACO	handheld	1,500	171 px	25
RoLID-11K	dashcam	11,564	**22 px**	84
UAVVaste	drone	772	72 px	66

and SHA-256 lineage; Figure 1 overviews the whole flow, from verified ingestion to the evaluation contract.

Table I summarizes the domains and Figure 2 shows the axis that separates them: object scale.

Audit findings. Systematic checks (perceptual hashing, EXIF-frame verification, split cross-referencing) surfaced seven defects we correct or document: (H1) TACO basenames collide across batches (92% of images share a filename with another batch); (H2) $\sim$37% of TACO images carry pending EXIF rotations, with boxes annotated on the *rotated* frame; (H3) RoLID's release folders do not encode capture hour; (H4) **RoLID's official splits leak at the video level**: 22 of 84 source videos span multiple splits and 58.2% of test frames share a video with training near-duplicate frames (87% visual redundancy by perceptual hash) make this scene leakage, inflating published baselines; (H5) one image is duplicated between official val and test; (H6) UAVVaste's official split also contains one near-duplicate pair across train/test; (H7) annotation frames are inconsistent across datasets TACO annotates on the EXIF-rotated frame while five UAVVaste handheld photos annotate the raw frame so pixels are materialized per-image in the frame that matches the declared dimensions, with EXIF metadata stripped.

Curation. Silver applies five corrections (duplicate removal from val, phantom-category removal, full-path identity, degenerate-box and MD5 flags), defines a two-level taxonomy *litter* (1 class, all datasets) and TACO material classes in two parallel variants, **A: 6 classes** (glass included, marginal at 38 projected test instances) and **B: 5 classes** (glass excluded) and freezes four split schemes with group-aware units: pHash visual groups for TACO/UAVVaste and *source video* for RoLID. TACO splits use iterative stratification [15]; every class lands within 0.1 pp of the 15% test target. For RoLID we keep *both* the official (leaky, for comparability) and a by-video (0% leakage) scheme, enabling a paired quantification of the inflation.

IV. Methods

P0 - fine-tuned classical pipeline. (1) *Faster R-CNN* (ResNet-50-FPN, COCO-pretrained) under two transfer strategies frozen backbone vs. unfreezing `layer4` with linear warmup, gradient clipping and early stopping on validation AP_{50}. (2) *FCN-ResNet18* for binary litter/background segmentation on TACO's rasterized polygons (1×1 conv + $\times32$ transposed conv with bilinear init, class-weighted cross-entropy). (3) *VGG-16 transfer learning* on 224-px litter/background crops (two strategies $\times$ two learning rates), providing the substrate for CAM [10], Grad-CAM [11] and Guided Backpropagation [12] interpretability panels. (4) An *anchor-coverage analysis*: FPN base anchors (32-512 px) against the measured object-size distributions, plus RPN objectness maps from the trained dashcam model a mechanistic account of the small-object gap.

P1 - modern reference. YOLOv11n [13] trained per domain at 640 px (and 1024 px for the resolution ablation), used to fill the 3×3 cross-domain matrix, the official-vs-by-video leakage pair, and the 6-vs-5-class variant comparison.

Evaluation contract. A single source of metric truth: predictions from every model are evaluated with COCO tooling against the same frozen ground truths; test is touched once per experiment with best-validation weights; saved predictions feed TIDE [14] error decomposition.

V. Experimental Setup

Nineteen runs (11 P0, 8 P1) executed on two lanes: Apple M-series (MPS) for VGG/FCN/YOLO and CUDA (Colab A100) for Faster R-CNN and the heavier YOLO runs profiling showed healthy Faster R-CNN iterations cost $\sim$95 s on MPS (variable input shapes force per-shape kernel recompilation), versus $\sim$0.3 s on CUDA; an hour-budget guardian projected total cost after the first epoch and routed unviable runs automatically. Budgets: generous epoch caps (50-150) with early stopping (patience 8-30) and validation curves; seeds fixed; per-epoch checkpoints with resume. Images are materialized at max side 1280 (Lanczos, JPEG q95) in the annotation frame (H7). Figure 3 shows the validation trajectories of the eight YOLO runs: clean convergence, early stopping at work, and the separation by domain and resolution.

Multiclass exports exclude *poisoned negatives* images whose only objects belong to excluded classes rather than training them as background (1/38/60 images for the 1/6/5-class variants).

VI. Results

Every number below is generated by the consolidated evaluation notebook from saved test predictions against the frozen ground truths; no metric is transcribed by hand.

A. Cross-domain matrix

Figure 4 presents the central result. In-domain, the three YOLOv11n models average 0.639 AP_{50}; across domains this collapses to 0.118 an **82% degradation**. The matrix is strongly asymmetric: the dashcam domain is an island ($AP_{50} \leq 0.064$ in either direction), while handheld$\rightarrow$drone transfers surprisingly well (0.527, retaining 68% of the drone in-domain score). This pattern indicts *object scale* rather than appearance alone: TACO and UAVVaste objects (median 171 and 72 px) both lie within the FPN anchor range, whereas RoLID's 22-px median falls below the smallest 32-px anchor consistent with our anchor-coverage analysis and with AP_S collapsing hardest (diagonal AP_S: 0.056 / 0.279 / 0.374).

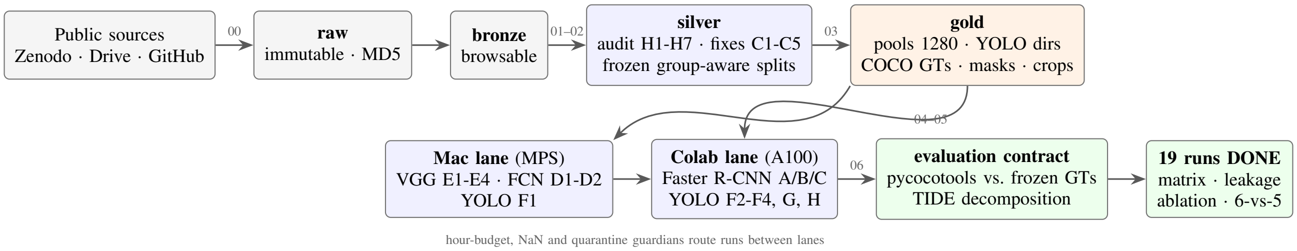

Fig. 1. End-to-end pipeline. Data flows through a medallion architecture with SHA-256 lineage (arrow labels: notebook numbers); training runs on two lanes governed by automatic guardians; every model is evaluated under a single metric contract.

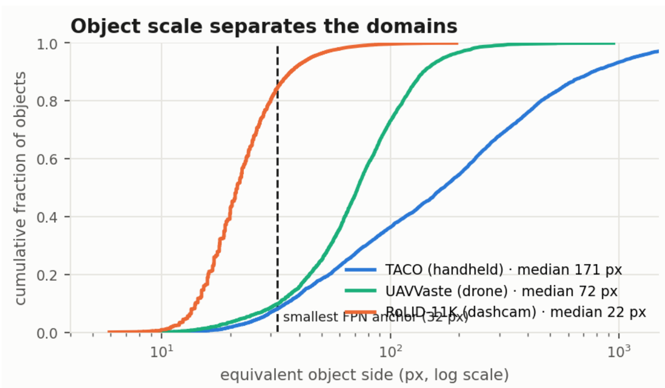

Fig. 2. Cumulative distribution of object side per domain. RoLID's median (22 px) falls below the smallest 32-px FPN anchor: the regime that explains the cross-domain matrix and motivates the resolution ablation.

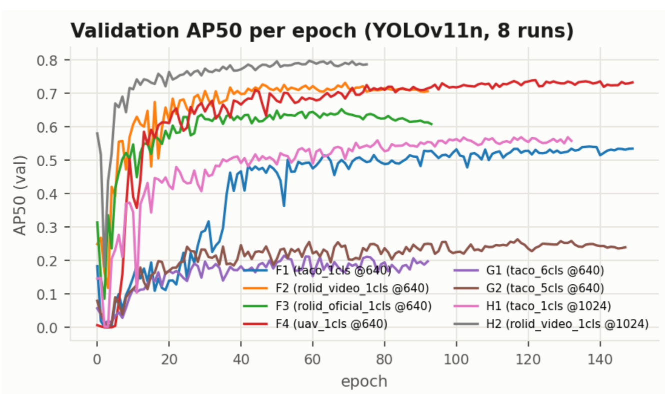

Fig. 3. Validation AP_{50} per epoch for the eight YOLOv11n runs.

B. Leakage inflation in RoLID-11K

Table II pairs the official (58% video leakage) and by-video (0%) protocols. On the *same* clean by-video test set, the model trained on the official split which saw other frames of those very test videos scores 0.844 AP_{50} versus 0.695 for the cleanly trained model: a **+21% relative inflation** attributable to scene memorization, not generalization (87% of intra-video frame pairs are visually redundant by perceptual hash). Published baselines on the official protocol should be read with this bias in mind; group-aware splitting removes it at zero cost.

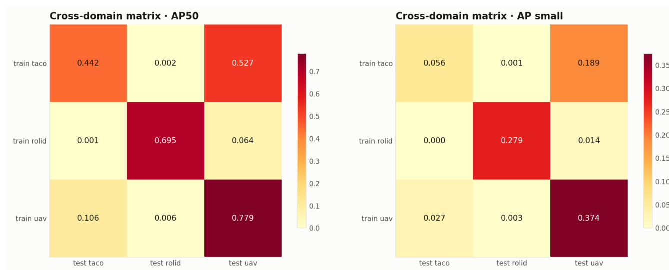

Fig. 4. Cross-domain matrix on test (YOLOv11n @640): AP_{50} (left) and AP_S (right). Rows: training domain; columns: test domain. In-domain mean 0.639 vs. cross-domain 0.118.

TABLE II
LEAKAGE QUANTIFICATION ON ROLID-11K (TEST AP_{50}, YOLOV11N @640).

Training split	Test set	AP_{50}
official (58% leak)	official	0.620
by-video (0% leak)	by-video	0.695
official (58% leak)	by-video	**0.844**

C. Families under the same protocol

Table III compares Faster R-CNN (41.3 M parameters) and YOLOv11n (2.6 M) on identical splits and ground truths. On single-class tasks the 16× smaller model stays within 1-4 AP_{50} points (0.695 vs. 0.707 on dashcam; 0.442 vs. 0.483 on handheld), supporting lightweight deployment; on material classification the gap widens to 5-9 points. Unfreezing `layer4` lifts Faster R-CNN by +3.7 AP_{50} over the frozen backbone (0.483 vs. 0.446). Our 5-class YOLO (0.192 mAP_{50}) is consistent with TrashDet's 19.5 on a comparable TACO subset [7], confirming that absolute scores in this problem are structurally low. The complementary P0 models complete the picture: FCN-ResNet18 reaches 0.577 test IoU on binary litter segmentation, and the VGG-16 crop classifier attains 97.8% test accuracy, providing the substrate for the CAM/Grad-CAM evidence that the network attends to the object rather than context (Figure 5).

D. Resolution and class granularity

Table IV shows the 640→1024 ablation. Gains concentrate exactly where objects are smallest: +8.3 AP_{50} and +5.8 AP_S on dashcam, versus +3.2 and +3.9 on handheld resolution is

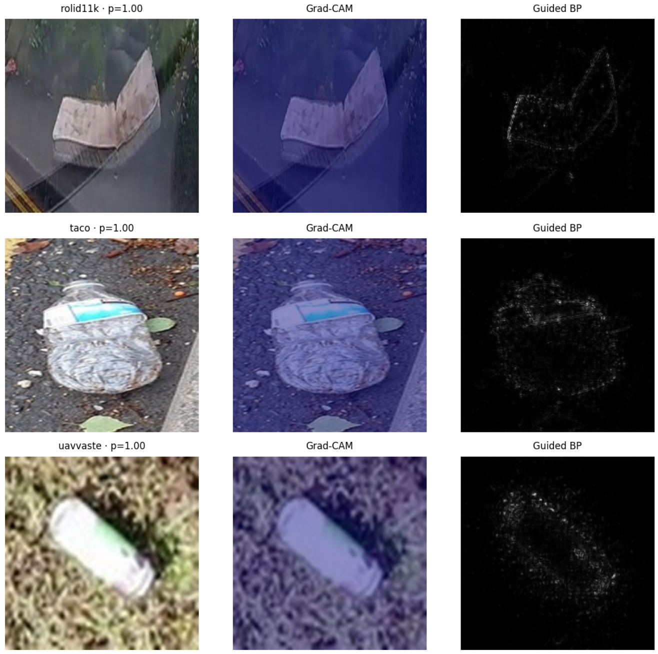

Fig. 5. Crop-classifier interpretability: one row per domain (dashcam, handheld, drone) with Grad-CAM and Guided Backpropagation. Activation concentrates on the object even in 22-px crops.

TABLE III
MODEL FAMILIES ON IDENTICAL SPLITS (TEST METRICS). FRCNN = FASTER R-CNN R50-FPN, `LAYER4` UNFROZEN.

Data	Model	AP_{50}	AP	AP_S
TACO 1-cls	FRCNN	**0.483**	**0.323**	**0.073**
	YOLOv11n	0.442	0.318	0.056
RoLID by-video	FRCNN	**0.707**	0.277	**0.281**
	YOLOv11n	0.695	**0.279**	0.279
TACO 6-cls	FRCNN	**0.246**	0.154	0.029
	YOLOv11n	0.157	-	-
TACO 5-cls	FRCNN	**0.244**	0.155	0.024
	YOLOv11n	0.192	-	-

the cheapest effective lever for the small-object regime. For class granularity, excluding the marginal glass class (38 test instances, AP_{50} 0.049 when kept) improves the global 5-class score to 0.192 vs. 0.157 for 6 classes, with a mean +0.014 AP_{50} on the five shared classes (largest on metal, +0.077): keeping a severely under-represented class taxes its siblings.

E. Error decomposition

Figure 6 decomposes the error with TIDE across all 13 detectors. Three regimes emerge. Multiclass models are dominated by *classification* error (up to 24.6 dAP in the 5-class YOLO, against only 1.7 for localization): finding litter is not the bottleneck naming its material is. Single-class handheld models split their error between background false positives and missed objects (~11-13 dAP each). In-domain dashcam models, counter-intuitively, miss little ($dAP_{Miss} \leq 3.9$): the

TABLE IV
RESOLUTION ABLATION (YOLOV11N, TEST).

Domain	AP_{50} 640	AP_{50} 1024	AP_S 640	AP_S 1024
TACO	0.442	**0.474**	0.056	**0.096**
RoLID	0.695	**0.779**	0.279	**0.337**

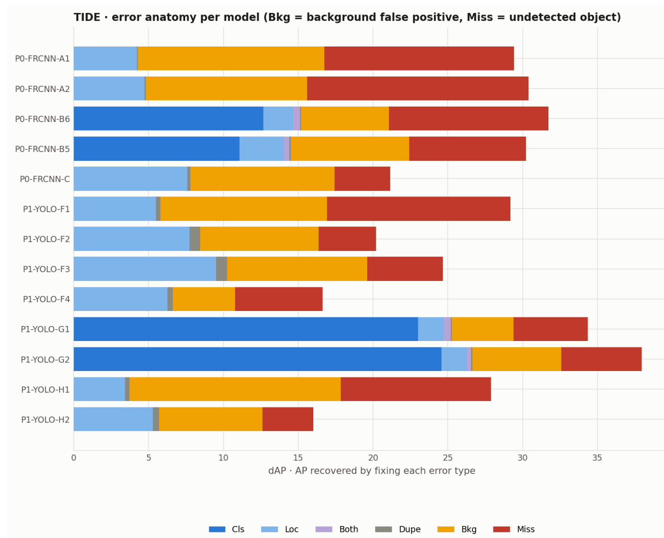

Fig. 6. TIDE error decomposition per model (dAP recovered by fixing each error type). Cls dominates multiclass models; Bkg+Miss dominate single-class ones.

22-px objects *are* found, and the residual error is dominated by background confusions evidence that the cross-domain collapse of Figure 4 stems from representations, not from an unrecoverable recall floor.

F. Qualitative out-of-distribution results

Figure 7 shows the batched demo (notebook 07) on a session of ten images downloaded from the internet, unrelated to the three datasets. Under conservative thresholds (0.25 for YOLO, 0.5 for Faster R-CNN), the quantitative patterns reappear out of distribution: Faster R-CNN proposes more candidates than YOLOv11n on all ten images (211 vs. 87 detections in total; 45 vs. 15 on the densest scene), the FCN estimates 3-16% litter pixels per scene, and the VGG classifier confirms every crop with P(litter)≥0.99, Grad-CAM attending to the object rather than its context. These images are illustrative evidence, not a metric: they carry no annotations.

VII. DISCUSSION AND LIMITATIONS

Three findings organize the evidence. **Litter detectors do not travel.** The 82% cross-domain drop, with dashcam isolated and handheld↔ drone partially compatible, is explained primarily by object scale: domains whose object sizes fall within the pretraining anchor range exchange some knowledge; the 22-px dashcam regime exchanges none. For municipal deployments mixing sensors, per-domain fine-tuning

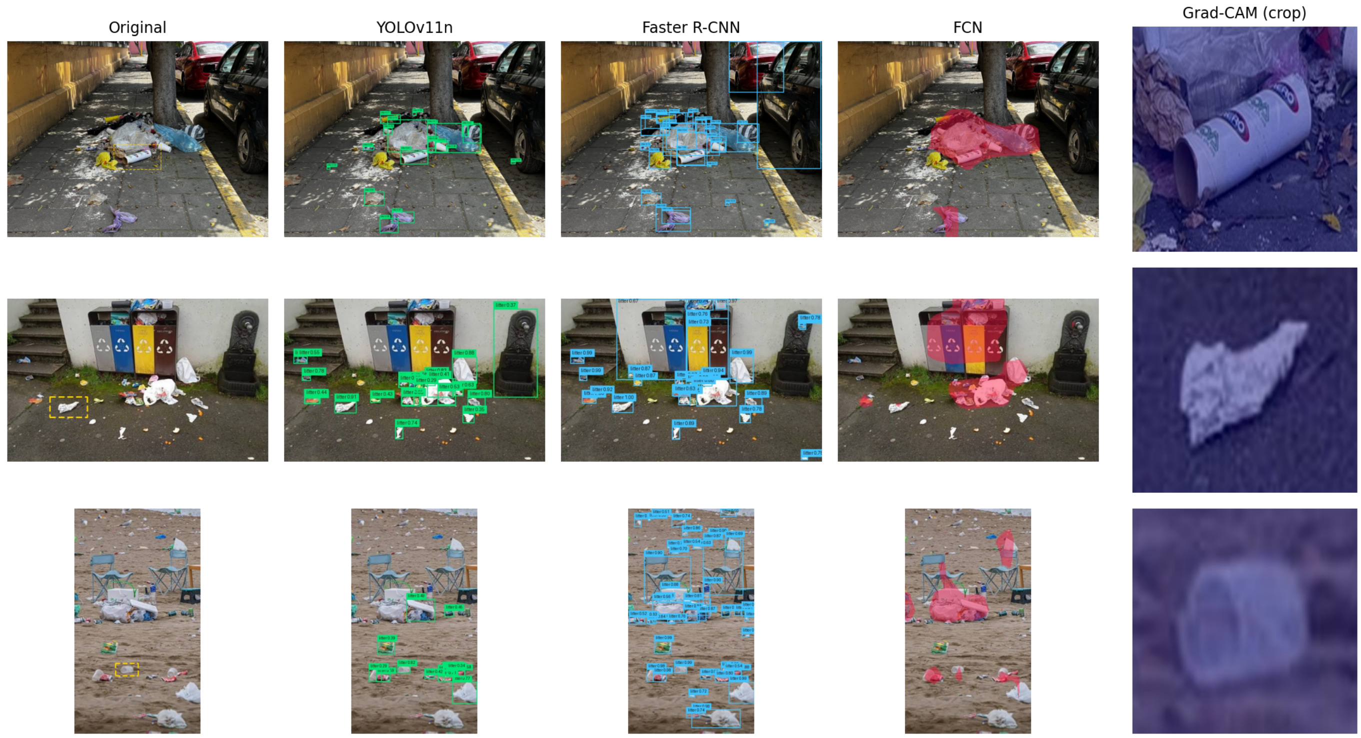

Fig. 7. Qualitative demo on out-of-distribution web images (three of the ten in the session): original with the Grad-CAM crop origin marked with a dashed box, YOLOv11n and Faster R-CNN detections, FCN litter mask and the crop classifier's Grad-CAM explanation. Images randomly downloaded from the internet, with no embedded authorship metadata, used for illustration only; they will be replaced by our own captures (Lima) in a future revision.

or at minimum resolution adaptation remains mandatory; a single universal litter model is not yet realistic. **Split hygiene changes conclusions.** The official RoLID protocol rewards scene memorization (+21% AP_{50}); results published on it are optimistic by roughly that margin, and the fix (group-aware, by-video splitting) costs nothing. We argue dataset releases with video or burst structure should ship group identifiers. **The cheap levers work.** 1024-px input buys back +8.3 AP_{50} on dashcam; a 2.6 M-parameter detector performs within a few points of a 41.3 M one on single-class litter detection; and TIDE locates the multiclass bottleneck in material classification rather than localization suggesting a decoupled design (class-agnostic detector plus crop classifier, where our VGG substrate reaches 97.8% accuracy) over ever finer detection heads. Limitations we can already state: absolute mAP in this problem is structurally low [7]; our compute budget precluded transformer detectors (CO-DETR-class) which lead RoLID's in-domain benchmark; StreetView-Waste could not be included (endpoints unavailable); our own OOD capture (Lima) remains future work, as does temporal aggregation over dashcam video natural given the video-level structure our audit exposed.

VIII. Conclusion

We audited three public litter datasets, released corrected group-aware splits, and measured under a single evaluation contract how far litter detectors travel across handheld, dashcam and drone imagery. The answers are concrete: an 82% mAP_{50} drop out of domain with dashcam fully isolated; a +21% AP_{50} inflation hiding in RoLID-11K's official splits; up to +8.3 AP_{50} recoverable by input resolution alone; and a 16$\times$ smaller modern detector within a few points of Faster R-CNN on single-class litter. Error decomposition shows the remaining battles are background discrimination and material classification, not localization. The full pipeline from checksummed ingestion to generated tables plus weights and a live demo are public; transformer detectors, temporal aggregation over dashcam video and our planned Lima out-of-distribution capture are the natural next steps.

Reproducibility

Code, notebooks (ingest to evaluation), audited splits and trained weights are released; three reproduction routes are documented (demo with pretrained weights, metric regeneration from released artifacts in $\sim$10 minutes, and full retraining from public data). Datasets are downloaded from their official sources with checksum verification.

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